Farhan Ahmed Onu

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Portfolio

Summary

CSE graduate and ongoing MSc student at Jahangirnagar University, focused on deep learning, efficient/quantized models for edge deployment, medical image analysis, and explainable AI. Published in IEEE venues and experienced with PyTorch, TensorFlow, ONNX Runtime, ROS2, and backend/data pipelines. IELTS Academic: 7.5.

Education

Master of Science in Computer Science & Engineering — Ongoing
Jahangirnagar University, Bangladesh — **Expected: November 2026**

Bachelor of Science in Computer Science & Engineering — CGPA: 3.4/4.0
Jahangirnagar University, Bangladesh — **2025**

Higher Secondary Certificate — GPA: 5.0/5.0
Jahangirnagar University School and College, Bangladesh — **2019**

Secondary School Certificate — GPA: 5.0/5.0
Jahangirnagar University School and College, Bangladesh — **2017**

Publications

- F. A. Onu, **Brain Tumor MRI Classification using Enhanced DenseNet169**, in *2026 IEEE QPAIN* . https://doi.org/10.1109/QPAIN69676.2026.11546223

  Developed a robust deep learning framework for classifying brain tumors (Glioma, Meningioma, Normal, Pituitary) from MRI scans. Achieved 98.60% accuracy using enhanced DenseNet169 with 5-fold cross-validation, custom data augmentation, and Grad-CAM explainability, with perfect classification of Pituitary tumors.

- F. A. Onu, **Latency-Aware Benchmarking of Deep Learning Models for RF Signal Classification**, in *2026 IEEE International Conference on Microwave, Antennas, RF and Sensors (MARSCON)*, accepted, in press.

  Benchmarked five deep learning architectures (DenseNet169, MobileNetV3, EfficientNet-B0, ViT-Small, and MobileViT-S) for RF signal classification (LTE, 5G NR, LTE-NR) under realistic impairments. Evaluated latency, throughput, peak memory, and FLOPs to identify optimal models for GPU, CPU, and memory-constrained edge deployments.

- F. A. Onu, S. M., J. K. Das, **Exploring Metabolic Subgroups and Algorithmic Fairness in a Bangladeshi Type-2 Diabetes Cohort: An Explainable Machine Learning Case Study**, in *IEEE iCONEECT*, accepted, in press.

  Applied explainable machine learning to a Bangladeshi Type-2 diabetes cohort to identify metabolic subgroups and assess algorithmic fairness across them, examining how subgroup-specific model behaviour may introduce bias in clinical risk prediction. This was a part of BSc research project.

Language Proficiency

IELTS Academic Overall Band Score: **7.5**

Listening **8.0** | Reading **8.0** | Writing **6.5** | Speaking **7.0**

Technical Skills

- **AI/ML & Data Science:** PyTorch, TensorFlow, Scikit-learn, LangChain, LLM Integration, Time-Series Forecasting, Diffusion Models, xLSTM (familiar), Autoregressive Modeling, Model Quantization (INT8/INT4), ONNX Runtime, Data Preprocessing, Model Evaluation
- **Programming:** Python, SQL, Java, C, C++, JavaScript/TypeScript
- **Robotics & 3D Perception:** ROS2, Open3D, Point Cloud Processing, Sensor Fusion, Monocular Depth Estimation
- **Backend & Data:** Django, Node.js, Express.js, RESTful APIs, ETL Pipelines
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite
- **Cloud & DevOps:** AWS, Docker, Git, GitHub Actions, CI/CD Pipelines, Linux
- **Other Tools:** Postman, Jira, Trello, Agile/Scrum

Other Projects

3D Mapping Using a Mechanically Oscillating 2D LiDAR
Built a sub-$150 3D scanner by mounting an RPLIDAR C1 on an oscillating MG996R servo, tagging each 2D scan with its tilt angle and reassembling slices into ROS2 PointCloud2 maps. Compared servo-commanded (open-loop) and MPU6050 IMU-assisted (closed-loop) tilt estimation, achieving 9.1 mm plane-fit RMSE on a flat-wall test; extended the pipeline with Poisson surface meshing and a monocular depth (Depth-Anything-V2) baseline for cross-modal comparison.
Showcased at Bangladesh Innovation Fair 2026.

Toward On-Device Mosquito Larvae Identification: Lightweight Blur-Robust Classifiers with INT8/INT4 Quantization
Trained MobileNetV3-Small, EfficientNet-Lite0, and MobileViT-S on the MosquitoLarvae-7400 benchmark (7,463 images, 4 species) using a two-stage regimen and a blur/motion-blur/color-jitter augmentation pipeline. Exported each model to ONNX and evaluated FP32, static INT8, and weight-only INT4 through real ONNX Runtime sessions; EfficientNet-Lite0 gave the best accuracy-robustness trade-off (99.73% clean / 98.66% blurred) and was the only model viable under static INT8 (96.50% at 3.5× compression). Benchmarked all variants end-to-end on PC and Android via onnxruntime-web WASM.

AI-Based Diabetes Diagnosis System
Developed an end-to-end AI system for diabetes diagnosis using machine learning and large language models (LLMs). The solution integrates predictive modeling with conversational AI for comprehensive patient assessment. The system is light-weight and can run on a budget GPU with 8GB VRAM. A portion of this project was accepted as a paper.

Localized RAG Pipeline for Cisco IOS Configuration Generation
Designed an offline RAG pipeline generating Cisco IOS CLI commands from natural language using ChromaDB and sentence-transformer embeddings. Created a validated dataset of 2,254 QA pairs and benchmarked seven small language models (Mistral, Qwen3, Gemma) using custom metrics. Mistral achieved a 51% exact match, demonstrating practical, privacy-preserving local network automation.

Volunteer Experience

Mentor & Member
Jahangirnagar University Robotics Club, Bangladesh **2023–2026**

References

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